

Tech Trends Shaping Micro-Mobility

The concept of micro-mobility was best displayed in Segway. As this field of innovation expands its borders to two-wheeler and three-wheeler options, will the market accept this change? What is the technology that's driving them?

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The demand for shared mobility has increased over the past decades across various cities all over the world. Rather than just being used for recreational purposes, shared electric two-wheelers and bicycles have become an increasingly popular means of daily transport.

This concept of shared mobility has the potential to help local governments deliver clean, healthy, and efficient urban mobility. So they have been launched in many European and American cities and can now be seen in some Indian cities like New Delhi as well.

However, the challenge with two-wheeler and three-wheeler micro-mobility vehicles is that they do not utilise the parked time for charging when left on sidewalks and other public spaces. If the vehicles are charged while parked, it is possible to reduce the size of their batteries and increase their life as the batteries would be trickle charged in small batches rather than having a deep charge every night.

Wireless charging

Wireless charging technologies are usually preferred to charge micro-mobility vehicles as they eliminate the need to plug in. Two types of wireless charging technologies exist, each having its own merits and demerits.

One is the Close Coupling technology in which the primary coil radiates energy and the secondary coil intercepts most of the magnetic flux. The separation, alignment, and sizes of the respective coils determine the coupling factor, which has a significant influence on the efficiency of the energy transfer. The frequency range of these systems is quite unique, as they start at high frequencies with light load and lower frequencies with higher load. This method provides high efficiency over changes in load.

The other technology is known as Resonant Inductive technology. The oscillating magnetic field generated by the primary coil induces a current in the secondary but also takes the advantage of the strong coupling

that occurs between the resonant coils—even when separated by tens of centimetres. These systems have a sharp resonant frequency peak. The optimum frequency, however, can change with load, separation, and presence of extraneous objects. Therefore, these systems require active frequency compensation in the transmitter, with more complex control and communication between transmitter and receiver. These systems may also cause electromagnetic interference for nearby electronic devices.

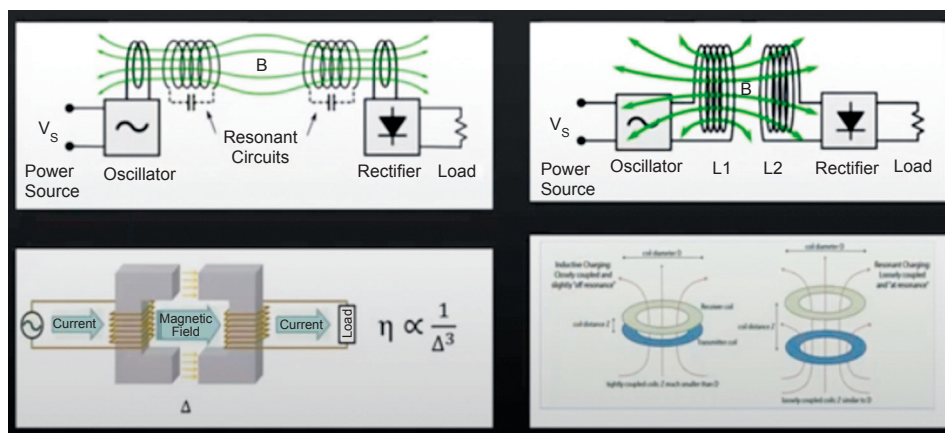


Fig. 1: Left side of the image shows a resonant system and right side shows a closely-coupled inductive wireless charging system